1. Basics of Engineering Economics

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1.1 Definition of Economics:

→ Economics is the science that deals with the production distribution and consumption of goods and services or the material welfare of human kind.

→ Economics can be broken into two main disciplines:

i) Macroeconomics:

→ It deals with the behavior of ~~large~~ economics on a large scale, usually the economics of countries or regions.

ii) Microeconomics:

→ It deals with the behavior of economics on a small scale, usually the economics of individual agent or person.

1.2 Demand:

→ Demand in economics is the consumer's desire and ability to purchase a goods or service.

1.2.1. Law of Demand:

→ Law of Demand states that demand is inversely proportional to the price of goods and services if other things remaining constant.

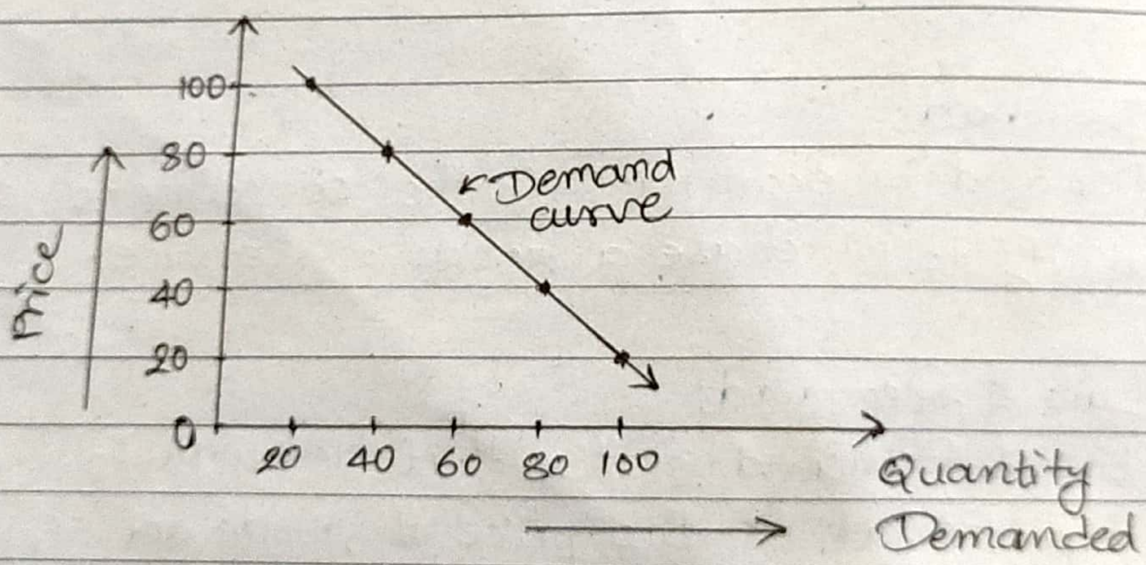
i.e. $\text{Demand} \propto \frac{1}{\text{Price}}$

Let us consider an example ~~of~~ of apple.

S.N.	Quantity Demanded (in kg)	Price/kg (in Rs)
1.	100	20
2.	80	40
3.	60	60
4.	40	80
5.	20	100

Fig. Demand Schedule

Representing above schedule in graph.



→ Figure above shows the demand curve which is downward sloping.

→ The higher the price of goods, lower will be the demand and vice-versa.

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1.2.2. Factor affecting Demand

i> Income of the people
ii> Taste of consumers
iii> Prices of related goods
iv> Custom and fashions
v> Weather
vi> Future expectations
vii> Price of Commodity

1.3. Supply:

→ Supply is defined as the desire with ability to sell and willingness to sell.
→ Supply is a fundamental economic concept that describes the total amount of a specific goods or service that is available to customers (consumers).

1.3.1. Law of Supply:

→ Law of Supply states that the supply is directly proportional to the price of goods and services if other things remaining constant.

i.e. Supply $\propto$ Price

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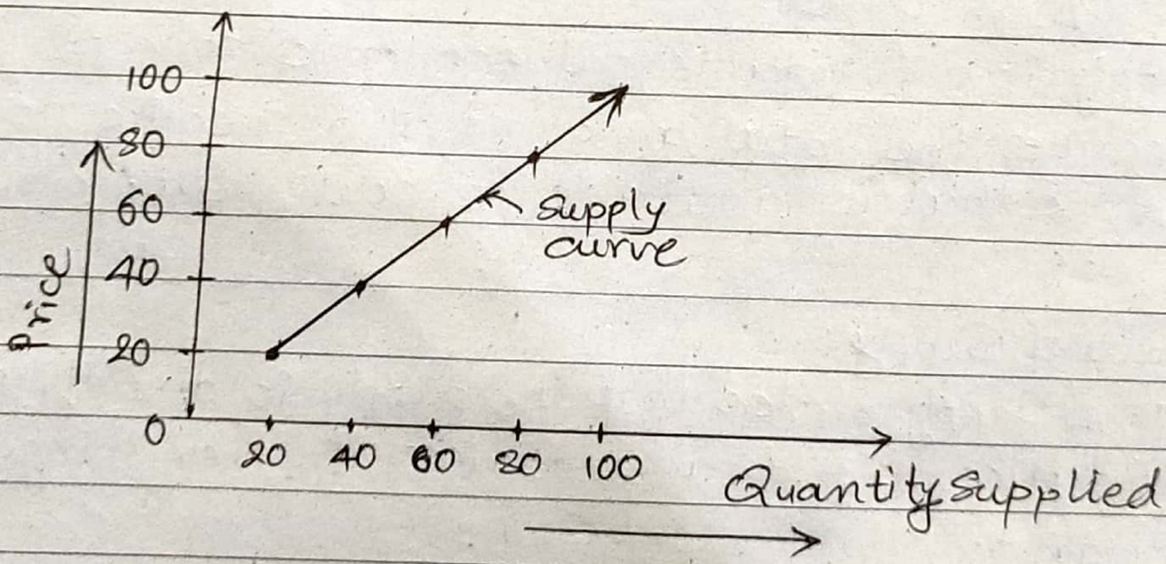
Let us consider an example of apple.

S.N.	Quantity Supplied (in kg)	Price/kg (in Rs.)
1.	20	20
2.	40	40
3.	60	60
4.	80	80
5.	100	100

Fig. Supply Schedule

Representing above schedule in the graph.



→ Figure above shows the supply curve which is upward sloping.

→ The higher the price of goods, higher will be the supply and vice-versa.

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1.3.2. Factors affecting Supply:

i> New Inventions
ii> Price of related goods
iii> Taxes and subsides
iv> Weather
v> Price of commodity
vi> Production technology
vii> Change in money income.
viii> State of natural resources.

1.4. Law of Demand:

→ Law of demand states that the demand is directly proportional to the supply if other things remaining constant.

i.e. Demand $\propto$ Supply

→ An interesting aspect of economy is that the demand and supply of a product are interdependent and they are sensitive with respect to the price of that product.

→ The inter-relationships between them is shown below.

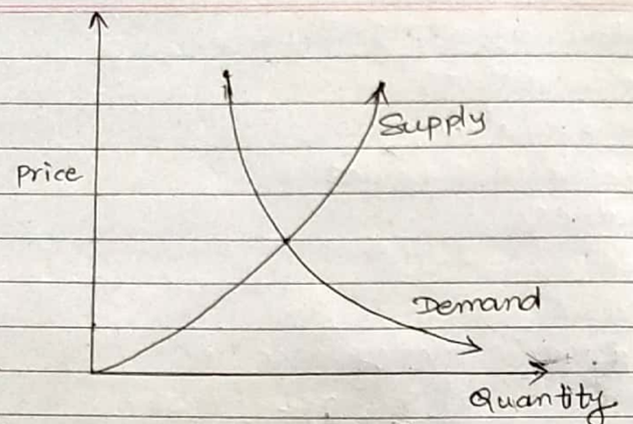


Fig. Demand & supply Curve

→ From above figure, it is clear that when there is decrease in the price of a product, the demand for the product increases and its supply decreases and vice-versa.

→ At the same time, lowering of the price of the product makes the producers restrain from releasing more quantities of the product in the market. Hence, the supply of the product is decreased.

→ The point of intersection of the supply curve and the demand curve is known as equilibrium point.

→ At the price corresponding to this point, the quantitity of the supply is equal to the quantity of demand. Hence, this point is equilibrium point.

1.5. Elasticity of Demand (E_d):

→ Elasticity of demand is the ratio of proportionate change in quantity demanded to the proportionate change in any one quantitative determinant of demand.

$$\text{i.e. } E_d = \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in any one quantitative determinant of demand}}$$

→ The elasticity of demand is classified as:

i> Price elasticity of demand (E_p)

ii> Income elasticity of demand (E_Y)

iii> Cross elasticity of demand (E_c)

1.5.1. Price Elasticity of Demand (E_p):

→ Price elasticity of demand is the ratio of proportionate change in quantity demanded of the commodity to the proportionate change in the price of the commodity.

$$\text{i.e. } E_p = \frac{\text{Proportionate change in quantity demanded to the commodity}}{\text{Proportionate change in the price of the commodity}}$$

$$\Rightarrow E_p = \frac{\Delta Q / Q}{\Delta P / P}$$

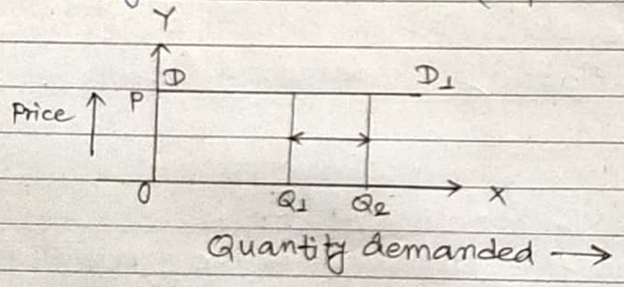
$$\therefore E_p = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

where,

ΔQ = change in quantity demanded
ΔP = change in price
P = Initial price
Q = Initial quantity demanded

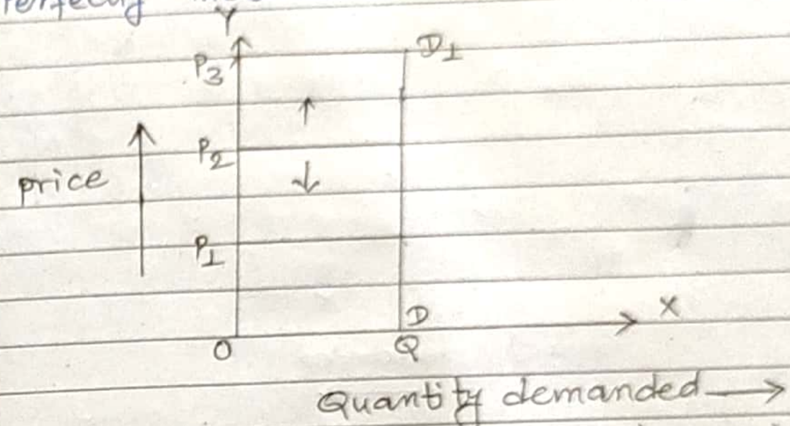
Degree of price ~~&~~ elasticity of demand:

1. Perfectly elastic demand ($E_p = \infty$).



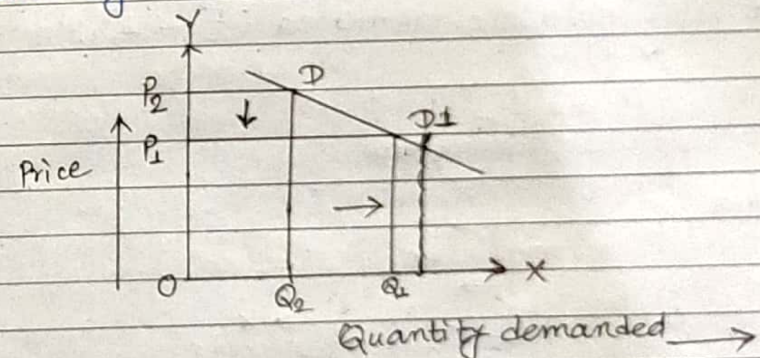
→ In the above figure, the negligible change in price causes infinite fall or rise in quantity demanded.

2. Perfectly inelastic demand ($E_p = 0$):



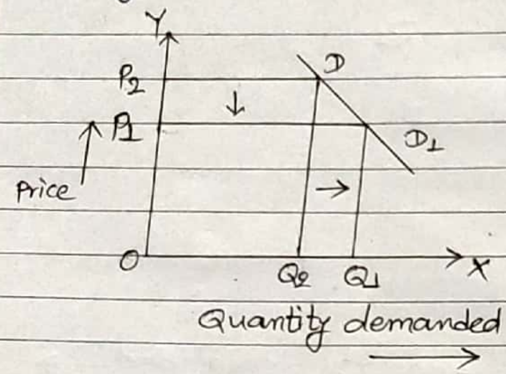
→ In the above figure, the demand remains unchanged irrespective of change in the price.

3. Relatively elastic demand ($E_p > 1$):



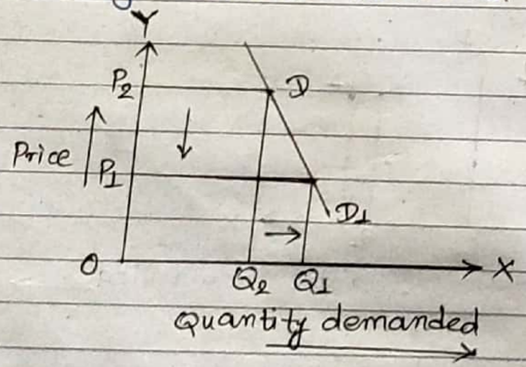
→ In the above figure, the demand curve is gradually slopping which shows that the proportionate change in quantity is greater than the proportionate change in the price.

4. Unitary elastic demand ($E_p = 1$):



→ In the above figure, the proportionate change in demanded quantity is equal to the proportionate change in the price.

5. Relatively inelastic demand ($E_p < 1$):



→ In the above figure, the demand curve is rapidly sloping which shows that the change in the quantity is smaller than the proportionate change in price.

1.5.2. Income elasticity of demand (E_r):

→ Income elasticity of demand is the ratio of proportionate change in quantity demanded to the proportionate change in income of consumer.

i.e. $E_r = \frac{\text{Proportionate change in quantity demanded}}{\text{Proportionate change in income}}$

$$\Rightarrow E_r = \frac{\Delta Q / Q}{\Delta Y / Y}$$

$$\therefore E_r = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

where,

ΔQ = change in quantity demanded

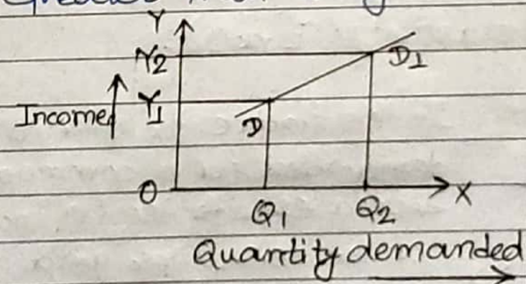
ΔY = change in income

Y = Initial income

Q = Initial demand of quantity

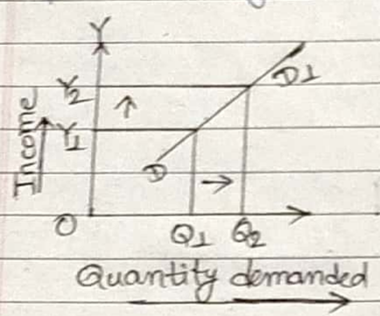
Degree of income elasticity of demand:

1. Greater than unity income elasticity ($E_r > 1$):



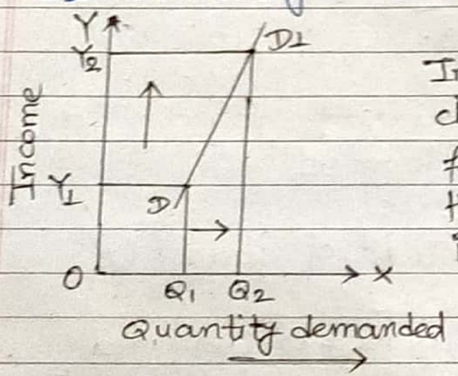
In this ~~abov~~ figure, the percentage change in quantity demanded for a commodity is greater than the percentage change in income of the customer.

2. Equal to unity income elasticity ($E_Y = 1$):



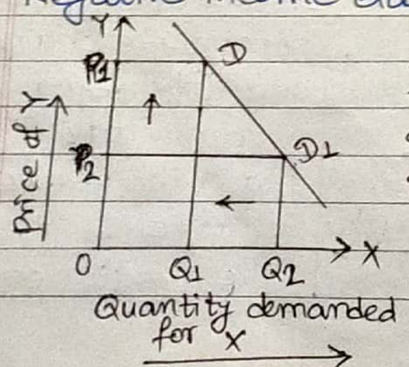
In this figure, the percentage change in quantity demanded for a commodity is equal to the percentage change in income of the consumer.

3. Less than unity income elasticity ($E_Y < 1$):



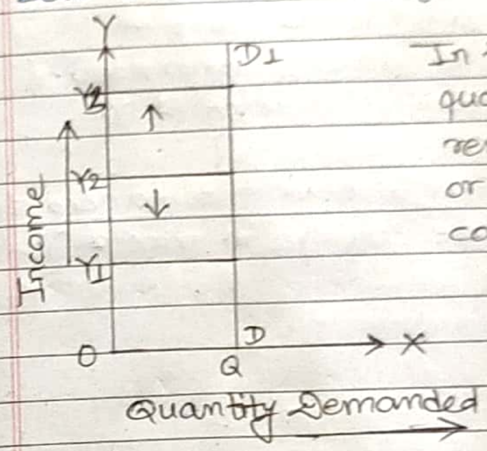
In this figure, the percentage change in quantity demanded for a commodity is less than the percentage change in income of the customer.
(consumer)

4. Negative income elasticity ($E_Y < 0$):



In this figure, the quantity demanded for a commodity decreases with rise in income of the consumer and vice-versa.

4. Zero income elasticity ($E_Y = 0$):



In this figure the ~~percentage~~ quantity demanded for a commodity remains constant with any rise or fall in income of the ~~co~~ consumer.

1.5.3. Cross Elasticity of Demand (E_c):

→ Cross elasticity of demand is the ratio of proportionate change in quantity demanded of X to the proportionate change in price of Y.

i.e.

$$E_c = \frac{\text{Proportionate change in quantity demanded of X}}{\text{Proportionate change in price of Y}}$$

$$\Rightarrow E_c = \frac{\Delta Q_X / Q_X}{\Delta P_Y / P_Y}$$

$$\therefore E_c = \frac{\Delta Q_X}{\Delta P_Y} \times \frac{P_Y}{Q_X}$$

where,

ΔQ_X = change in quantity demanded for X commodity

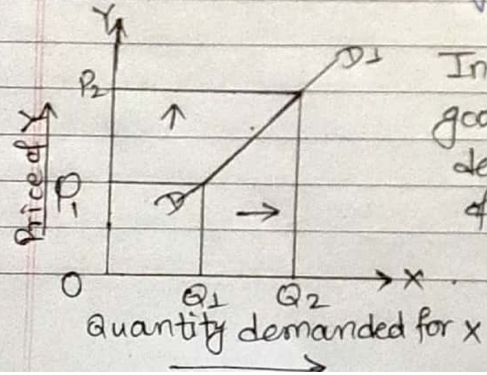
ΔP_Y = change in price of Y commodity

Q_X = Initial demand for X commodity

P_Y = Initial price of Y commodity

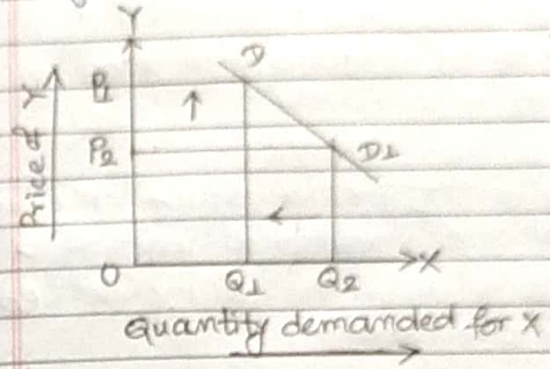
Degree of cross elasticity of demand:

1: Positive cross elasticity (~~of~~ E_c = +ve):



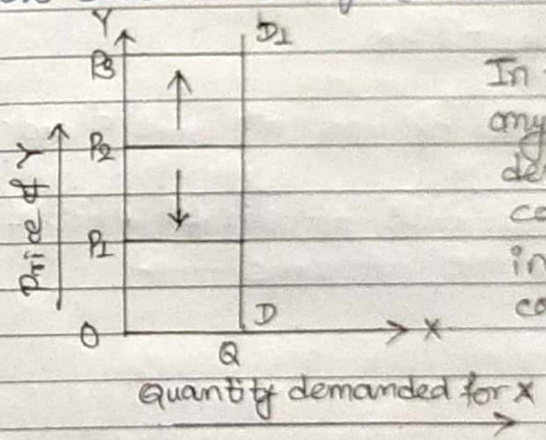
In this, the rise in price of goods leads to rise in quantity demanded ~~fo~~ of other goods of a similar nature and vice-versa.

2. Negative Cross elasticity ($E_c < 0$):



In this, the rise in price of one good leads to fall in quantity demanded of its complementary goods and vice-versa.

3. Zero cross elasticity ($E_c = 0$):



In this, there is no any response in quantity demanded for one commodity due to change in price of another commodity.

~~1.6. Law of Diminishing Marginal Utility~~ :

1.6. Utility and Marginal Utility :

→ Utility is the level of satisfaction to a consumer by consuming goods or services.

→ Marginal Utility is the extra utility that a consumer gets from one additional unit of a specific product. It is ~~deri~~ the utility derived by single unit of consumption.

$$MU = \frac{\text{change in total utility}}{\text{change in quantity consumption}}$$

$$\Rightarrow MU = TU_{N+1} - TU_N$$

where,

TU_N = Total utility up to n^{th} unit of outcome consume

TU_{N+1} = Total utility upto $(n+1)^{th}$ unit of outcome consume

1.7. Law of Diminishing Marginal Utility :

→ The law of diminishing marginal utility states the marginal utility gradually decreases with the leve of consumption utility being defined as benefit o satisfaction.

→ This law is based and explained under certain assumptions such as ; Utility is measurable, continuous consumption, Commodity is measurabl consumer taste and preferences unchanged etc

→ Let us consider the following table ;

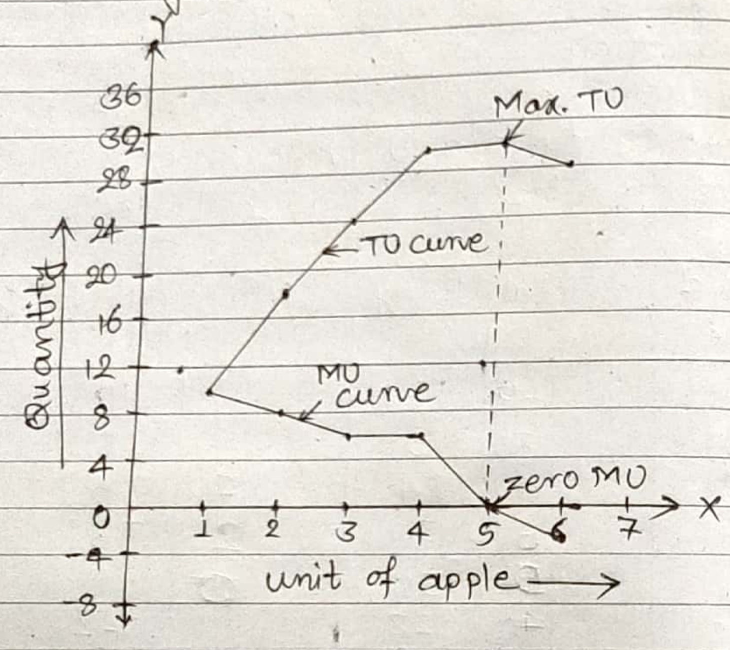
Unit of Apple	Total Utility (TU)	Marginal Utility (MU)
1	~~20~~ 10	~~20~~ 10
2	~~30~~ 18	~~10~~ 8
3	~~40~~ 24	~~10~~ 6
4	~~50~~ 30	~~10~~ 6
5	~~60~~ 30	~~[illegible]~~ 0
6	~~40~~ 28	-2

→ Suppose a person eats ~~bread or~~ apple and 1^{st} unit of apple gives him maximum satisfaction.

→ When he will eat 2^{nd} apple, his total satisfaction would increase.

→ But the utility added by 2^{nd} apple will be less than 1^{st} apple and this marginal utility will continue to decrease as he will keep on eating additional apple and once his stomach is full, marginal utility will become negative, decreasing the total utility.

→ Now, Plotting the above schedule;



In the above graph, the total utility (TU) curve is increasing at decreasing rate and it becomes maximum when marginal utility (MU) is zero.

1.8. Engineering Economics:

→ Engineering economics is the science that deals with the methods that enables one to take economic decisions towards minimizing cost and maximizing benefits to business organization.

→ Some special characteristics of engineering economics are as follows:-

i> Engineering economics is closely related to aligned with conventional micro economics.

ii> Engineering economics is devoted to the problem solving and decision making at the operation level.

iii> Engineering economics mainly uses the body of economic concepts and principles.

iv> Engineering economics integrates economic theory with engineering practices.

v> Engineering economics removes complicated abstract issues of economical theory.

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1.9. Principles of Engineering Economics:

→ Engineering economics principle focus on the process used to make an economics based, not on the decision itself.

→ There are several principle of engineering economics :-

i> Develop the alternatives:

→ There must be more than one alternatives, the choice is made among these after subsequent analysis.

ii> Focus on the differences:

→ Only the differences in expected outcomes of the feasible alternatives.

→ Be very neutral and impartial between the alternatives.

iii> Use a consistent viewpoint (economic viewpoint):

→ The prospective outcomes of the alternatives, selection of the criteria and other, should be consistently developed from a defined viewpoint.

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iv> Use a common unit of measure:

→ Using a common unit of measurement to enumerate as many of the prospective outcomes as possible will simplify the analysis of the alternatives.

v> Consider all relevant criteria:

→ Selection of a preferred alternative requires the use of a criteria or several criteria. For exampl long term interest.

vi> Make uncertainty explicit:

→ Risk and uncertainty are inherent in estimating the future outcomes of the alternatives and shoulc be recognized in the analysis and comparison.

vii> Revisit your decisions:

→ The initial projected outcomes of the selected alternative should be subsequently compared with the actual results achieved.

The first two principles setup the thought process.
The next three principles focus on evaluation criteria.
The last two principles focus on analysis.

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Applications of Engineering Economics :

→ The applications of engineering economics are diverse and found in most areas of an organizations.

→ They are typically classified as:-

i) Equipment and process selection.
ii) Equipment replacement
iii) New product and production expansion.
iv) Cost reduction
v) Capital Budget allocation
vi) New project evaluation etc

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2. Cost Concept and Fundamentals of Cost Accounting

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2.1. Cost terminology: Manufacturing and Non-manufacturing cost:

Cost:

→ Cost refers to the amount of expenditure incurred in acquiring something.

→ In general, cost refers to an amount to be paid or given up for acquiring any resources or services.

→ In economics, cost can be defined as monetary valuation of efforts, materials, resources, time and ~~utilies~~ utilities consumed, risk incurred and opportunity forgone in production of a good or service.

Elements of Cost:

1. Material Cost:

→ Material implies a substance from which a product is made. This include the cost of commodities and materials used by the organization.

i> Direct material Cost:

→ Direct material costs are those costs of materials that are used to produce the product.

→ Examples: cost of paper in book, wood in furniture, steel in bridge, gearbox in a car etc.

ii> Indirect material cost:

→ Indirect material costs are those costs of materials that are not used to produce the product.

2. Labour Cost:
→ Labour cost is the main element of cost.
→ An organization requires labour to convert raw materials into finished goods.

i) Direct labour cost:
→ Direct labour cost is the amount of wages paid to the direct labour involved in the production activities.
→ Examples: Welders in metal fabrication, carpenters in building works etc.

ii) Indirect Labour Cost:
→ Indirect labour cost is the amount that are not related to the direct labour.

3. Expenses:
→ Expenses refers to the cost that are incurred in production of finished goods other than material costs and labour costs.

i) Direct expenses:
→ Direct expenses are the production expenses that can be directly and conveniently traced to a unit of product.
→ Examples: Special designs and drawings, hire of special tools and equipment for manufacturing job and their maintenance.

ii) Indirect expenses:
→ Indirect expenses ~~that~~ are those expenses that are not directly traceable or those extremely small in monetary value.
→ Examples: Factory rent, lighting, insurance charges etc.

4. Overhead cost:
→ Overhead cost is the aggregate of indirect material cost, indirect labour cost and indirect expenses.

i) Manufacturing overheads:
→ Manufacturing overhead is the indirect expenses of operating manufacturing divisions of a concern and covers all indirect expenditure incurred.

ii) Administrative overhead:
→ Administrative overhead is the indirect ~~expenses~~ expenditure incurred in administrating the business.

iii) Sales overhead:
→ Sales overhead is the total expenses that is incurred in the promotional activities, marketing and selling activities excluding distribution.

iv> Distributed overhead:

→ Distributed overhead is the total cost of shipping the items from the factory site to the customer site.

2.1.1. Manufacturing Cost:

→ Manufacturing cost is the sum of all resources consumed in the process of making a product.

→ The manufacturing cost is defined as the costs that are incurred during the production of product.

→ Manufacturing costs includes direct material costs, direct labour costs and manufacturing overhead costs.

→ Manufacturing cost = cost of (labour + overhead + material

i> Direct material:

→ Milk for curd, cloth in pant, plastic for toys, wood for table etc.

ii> Direct Labour:

→ Salary for labour, salary for security guards, salary for supervisiors etc.

iii> Direct expenses:

→ Tax, depreciation, insurance, cost of patents, design cost etc.

2.1.2. Non-Manufacturing cost:

→ Non-manufacturing costs are the cost ~~of~~ for the company operations that are not directly related to manufacturing.

→ Non-manufacturing costs include marketing cost and administrative costs.

→ Marketing costs include all costs necessary to secure customer orders and get ~~finished~~ the finished product or service into hands of the customer.

→ For example: marketing costs ÷ secure customer order, shipping, commissions, advertisements etc.

→ For example: Administrative costs - public relatio[n] cost, management salaries, office supplies etc.

2.2. Cost for Business Decision:

2.2.1. Differential cost:

→ Differential cost is the difference between cost of two alternative decisions or of a change in output levels.

→ A differential cost can be a variable cost, a fixed cost or a mix of the two. There is no difference between these types of costs.

→ The differential cost is the difference in the prices paid plus the cost of operating the machines.

2.2.2. Differential Revenue:

→ Differential revenue is the difference in sales that will be generated by two different courses of action.

→ The concept is commonly used when evaluating which of two (or more) investments to make in a businesses.

→ For example;
You have a job paying Rs. 30,000 per month in your hometown. You have a job offer in neighboring city that pays Rs. 35,000 per month. The commuting cost to the city is Rs. 2,000.
Hence, Differential cost = Rs. 2,000
Differential revenue = Rs. 35,000 - Rs. 30,000
= Rs. 5000

2.2.3. Opportunity Cost:

→ The opportunity cost is the cost of sacrifing one opportunity.
Opportunity lost is the opportunity cost.

→ Opportunity

→ The loss of ~~another~~ other alternatives when one alternative is choosen.

→ The profit / lost when one alternative is choosen over another.

→ Example;

Project A	Project B
5 %	8 %
Accepted	×

i.e. Opportunity cost = 8 - 5 = 3 %

2.2.4. Sunk Cost:

→ Sunk cost is a cost that has already been incurred and cannot be recovered.

→ Sunk costs are not relevant to decisions because they cannot be changed regardless of what decision is made now or in the future.

→ Example; ~~You bought a bike~~ once a rent is paid, that dollars amount is no longer recoverable it is sunk.

2.2.5. Marginal Cost:

→ Marginal cost is defined as the added cost that would result from increasing the rate of output by a single unit.

→ $\text{Marginal cost} = \dfrac{\text{change in costs}}{\text{change in quantity}}$

→ Example;

current unit of production = 1000
current cost of production = \$100000
Future unit of production = 2000
Future cost of production = \$ 125000

$\therefore \text{Marginal Cost} = \dfrac{\$125000 - \$100000}{2000 - 1000}$

$= 25$ Ans

2.2.6. Fixed Cost:

→ Fixed costs are those costs that the firm has to pay independently of whether it is operating or not.

→ For example, rent of a building.

→ Fixed costs are expenses that have to be paid by a company, independent of any business activity.

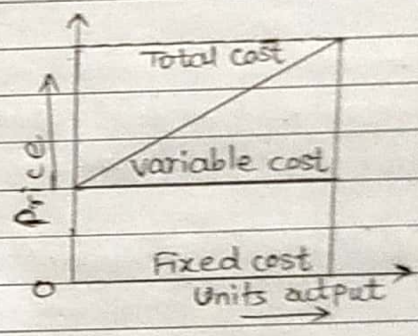
2.2.7. Variable Cost:

→ Variable costs are those cost that vary with the level of output.

→ For example: Wages paid to labour, cost of raw materials etc.

2.2.8. Total Cost:

→ Total cost is the summation of fixed cost and the variable cost.



2.2.9. Prime Costs:

→ The sum total of direct material, direct labour and direct expenses. For e.g. cost of raw materials.

2.2.10. Overhead Cost:

→ Overhead cost refers to an ongoing expense of operating a business.

→ For example: telephone bills, legal fees, supplies, repairs etc.

3. Time Value of Money

3.1. Interest:

→ Interest is the price of money services.

→ Interest is the fee that is charged for the use of money.

3.1.1. Simple Interest:

3.1.2 Compound Interest:

S.N.	Simple Interest	Compound Interest
i)	SI is calculated only on original principal.	CI is calculated on interest earned and the principal
ii)	SI is linear in nature.	CI is exponential in nature.
iii)	In SI, returns earned on the investment are lower.	In CI, returns earned on the investment are higher.
iv)	SI is smaller than CI.	CI is larger than SI.
v)	In SI, interest charged on principal.	In CI, interest charge on principal plus accumulated interest.
vi)	$SI = p * t * r$ where, P = Principal t = time period r = rate of interest.	$CI = P(1+r)^t - P$ where, P = Principal t = time period r = rate of interest
vii)	In SI, principal remains constant.	In CI, principal goes on changing during borrowing period.

3.1.3. Nominal Rate of Interest

3.1.4. Effective Rate of Interest

S.N.	Effective Interest Rate, i	Nominal Interest Rate, r
i)	Effective interest rate is the actual rate of interest earned during one year.	Nominal interest rate is the basic annual rate of interest.
ii)	It is greater than nominal interest rate.	It is smaller than effective interest rate.
iii)	It is often referred to as the annual percentage yield (APY).	Nominal interest rate is often referred to as the annual percentage rate (APR).
iv)	It can be used in the time value formulas or equations.	It cannot be used in the time value formulas or equations.
v)	It affects the real value of money one corresponding period.	It is convertible more than once in a corresponding period.
vi)	~~For example: For an interest rate of 5.6 % per month, the real interest rate is; i_{eff} $(1+0.056)^{12}-1=$~~ $i_{eff} = \left(1+\frac{r}{m}\right)^m - 1$	$r = N\left(\frac{T}{P}\right)^{\frac{1}{NY}} - N$

Time Value of Money:

→ Time value of money (TVM) is the fundamental concept in comparison of the economic performance between projects i.e. alternatives.

→ Time value of money is defined as the time dependent value of money because of changes in purchasing power and real time capacity of money over time.

→ Time value of money is the idea that money that is available at the present time is worth more than the same amount in the future, due to its potential capacity.

→ Time value of money is funded on time preference.

→ Time value of money is an important concept to the investors because a dollar on hand today is worth more than a dollar promised in the future.

→ When we deal large amount of money, long period of time or high interest rates, the change in the value of a sum of money over time becomes extremely significant.

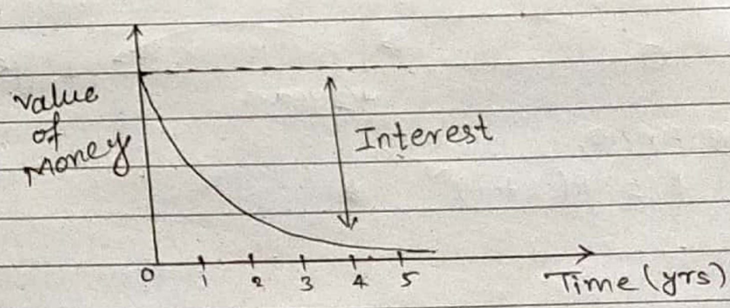
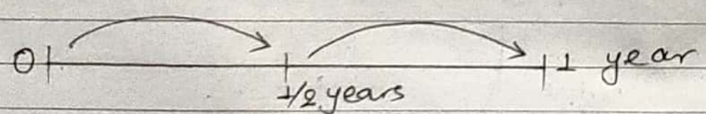


Fig. Time vs value of Money

Q. Why does bank use concept of compound interest instead of simple interest?

Sol:
→ Interest can compound either frequently (daily or monthly) or infrequently (quarterly, semi-annually or yearly).
→ The more often your interest compounds, the more interest ~~will~~ you will earn on your investment in compound interest rate.
→ When it comes to investing, compound interest is better since it allows funds to grow at a faster rate than they would in an account with a simple interest rate.
→ So, the bank uses concept of compound interest instead of simple interest.

Relationship between i & r:
→ Let us suppose, the interest rate is compounded semi-annually, i.e.



We know,

$$F = P(1+i)^N$$

Let i = effective interest rate per year
r = nominal interest rate per year
m = no. of compounding periods in year.

Then,
The interest rate per single compounding period would be $i = \frac{r}{m}$

We can also write,

$$P\left(1+\frac{r}{m}\right)^m = P(1+i)^1$$

$$\Rightarrow \left(1+\frac{r}{m}\right)^m = (1+i)$$

$$\therefore i = i_{eff} = \left(1+\frac{r}{m}\right)^m - 1$$

When $m = 1$, $i_{eff} = r$ then,

$$\left(1+\frac{r}{m}\right)^m = 1 + i_{eff}$$

$$\Rightarrow \left(1+\frac{r}{m}\right) = (1+i_{eff})^{1/m}$$

$$\Rightarrow \frac{r}{m} = (1+i_{eff})^{1/m} - 1$$

$$\therefore i_{monthly} = i_m = (1+i_{eff})^{1/m} - 1$$

3.2. Economic Equivalence:

→ Economic equivalence refers to a cash flow whether a single payment or a series of payments can be converted to an equivalent cash flow at any point in time.

→ Economic equivalence is commonly used in engineering field to compare alternatives (projects).

3.2.1. Present Worth (PW):

→ Present Worth of a given series of cash flow is the equivalent value/worth at the end of year zero or beginning of year 1.

3.2.2. Annual Worth (AW):

→ Annual Worth of a project is a uniform series of amount for a stated study period.

3.2.3. Future Worth (FW):

→ Future Worth of a project is the equivalence worth of all cash flows at the end of study period.

3.3. Cash Flow Diagrams:

→ Cash flow is the statement which shows inflows and outflows of cash and cash equivalents during life of projects.

→ The graphical representation of different cash flow streams is known as cash flow diagrams.

→ The cash flow diagram should show following things:

i) A time interval divided into an appropriate number of equal periods.

ii) All cash outflows in each period.

iii) All cash inflows in each period.

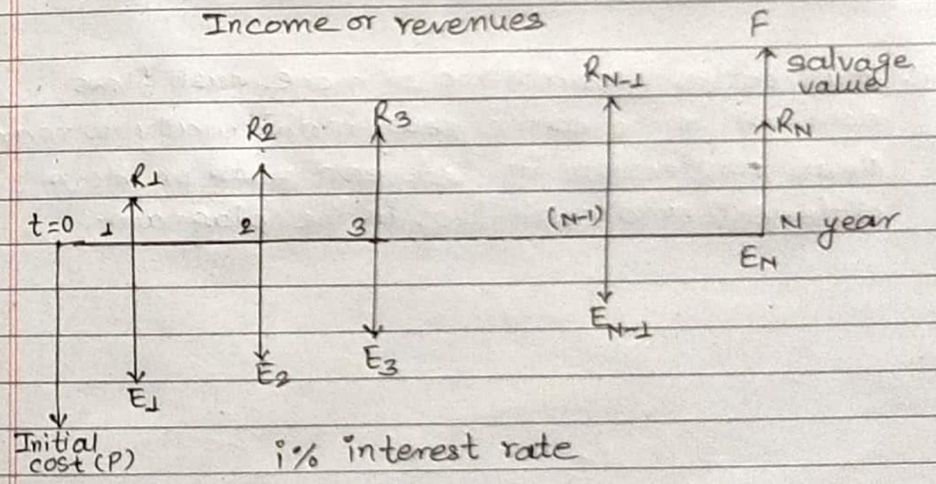


Fig. Cash Flow diagram

Q. How to draw cash flow diagram?

Sol To draw cashflow diagram, apply following steps:-

i> Horizontal line is time scale with ~~progress of~~ progression of time moving from left to right. (Normally starts from 0 to N years) where end of previous year coincide with beginning of next year.

ii> Inflows and Outflows of money is expressed by upward and downward arrows respectively. Here, inflows of cash represent annual benefit, salvage value etc and outflows of cash represent initial capital investment, annual cost etc.

iii> It is better to show two or more cash flows occuring in the same year individually to make clear connection in between ~~path~~ problem statement and cash flows in the diagram.

3.3.1. Types of Cashflows:-

1> Single cash flow:

i> Finding F when P is given:

$$F = P(1+i)^N$$

$$F = P\left(1+\frac{i}{m}\right)^{N\times m}$$

where,

F = Future Amount

P = Present Amount

i = interest rate

N = time period

m = no. of compounding period.

ii> Finding P when F is given:

$$P = \frac{F}{(1+i)^N}$$

$$P = \frac{F}{\left(1+\frac{i}{m}\right)^{N\times m}}$$

Book

Numerical:

(A) 1. What will be the maturity amount of Rs. 100000 after 5 years for nominal interest rate of 7% compounding quarterly?

Sol: Given;-

Present Amount, P = Rs. 100,000

Time Period, N = 5

Interest rate, $i = 7\% = 0.07$

No. of Compounding period, m = 4 i.e. quarterly

Future Amount, F = ?

We know,

$$F = P\left(1 + \frac{i}{m}\right)^{N \times m}$$

$$= 100,000 \left(1 + \frac{0.07}{4}\right)^{5 \times 4}$$

$\therefore F$ = Rs. 141,477.82

Hence, the maturity amount will be Rs. 141,477.82

Ans

2) Regular or Uniform Cashflow:

i) Finding F when A is given;

$$F = A\left(\frac{(1+i)^N - 1}{i}\right) \qquad A = \text{Regular Cash flow}$$

$$\text{or, } F = A\left(\frac{(1+i/m)^{N \times m} - 1}{i/m}\right)$$

ii) Finding P when A is given;

$$\boxed{P = A\left[\frac{(1+i)^N - 1}{i(1+i)^N}\right]}$$

$$\boxed{P = A\left[\frac{(1+i/m)^{N \times m} - 1}{i/m\,(1+i/m)^{N \times m}}\right]}$$

2018 - Fall

Book

Numericals:

(A5) 1) If you have Rs. 10,00,000 loan now from a bank, how much Rs. should you pay as installment per two month for 5 years if bank interest rate is 12 % per year?

Sol Given;

Present Amount, P = Rs. 10,00,000

Time Period, N = 5 years

Interest Rate, i = 12 % = 0.12

No. of Compounding period, m = 6

Equal installment, A = ?

We know,

$$P = A\left[\frac{(1+i/m)^{N\times m}-1}{i/m(1+i/m)^{N\times m}}\right]$$

$$\Rightarrow 10,00,000 = A\left[\frac{(1+0.12/6)^{5\times 6}-1}{0.12/6\,(1+0.12/6)^{5\times 6}}\right]$$

$$\Rightarrow 10,00000 = A \times 22.396$$

$$\Rightarrow \frac{10,00,000}{22.396} = A$$

$\therefore$ A = Rs. 44650.830

Hence, the required installment should be Rs. 44,650.830. Ans

2017 - Spring

Book

(A6) 2) You wish to study your son in medical college after 20 years. Recently, government has fixed total Rs. 35 lakh to complete MBBS studies. How much you need to deposit on each year to meet your desire? If bank providing 10 % interest rate per year for your fixed account?

Sol: Given:-

Future Amount, F = Rs. 35,00,000

Time period, N = 20 years

Interest rate, i = 10 % = 0.1

Equal Deposit, A = ?

We know,

$$F = A\left[\frac{(1+i)^{N}-1}{i}\right]$$

$$\Rightarrow 35,00,000 = A\left[\frac{(1+0.1)^{20}-1}{0.1}\right]$$

$$\Rightarrow 35,00,000 = A \times 57.275$$

$$\Rightarrow \frac{35,00,000}{57.275} = A$$

$\therefore$ A = Rs. 61108.686

Hence, I need to deposit Rs. 61, 108.686 on each year to meet my desire. Ans

Book single cash flow

2019-Spring

(47) ⇒ What is the interest rate if your amount will be double in 5 years?

Sol: Given;

Time period, $N = 5$ years

Interest rate, $i = ?$

Let Present amount $= x$

Future amount $= 2x$

we know,

$$F = P(1+i)^N$$

$$\Rightarrow 2x = x(1+i)^5$$

$$\Rightarrow 2 = (1+i)^5$$

$$\Rightarrow \ln(2) = 5\ln(1+i)$$

$$\Rightarrow \frac{0.693}{5} = \ln(1+i)$$

$$\Rightarrow 0.1386 = \ln(1+i)$$

$$\Rightarrow e^{0.1386} = 1+i$$

$$\Rightarrow i = e^{0.1386} - 1$$

$$\Rightarrow \therefore i = 0.1486$$

i.e. $i = 14.86\ \%$

Hence, the required interest rate is 14.86 %. Ans

3) Linear Gradient Series (Cashflow):

Let

G = Growth Amount or Gradient Amount

i) Finding F when G is given:

$$F = \frac{G}{i}\left[\frac{(1+i)^N - 1}{i}\right] - \frac{NG}{i}$$

or,

$$F = \frac{G}{i/m}\left[\frac{(1+i/m)^{N\times m} - 1}{i/m}\right] - \frac{NG}{i/m}$$

ii) Finding P when G is given:

$$P = \frac{G}{i}\left[\frac{(1+i)^N - 1 - Ni}{i(1+i)^N}\right]$$

$$\text{or } P = \frac{G}{i/m}\left[\frac{(1+i/m)^{N\times m} - 1 - N\times m\times i/m}{i/m\,(1+i/m)^{N\times m}}\right]$$

iii) Finding A when G is given:

$$A = G\left[\frac{1}{i} - \frac{N}{(1+i)^N - 1}\right]$$

$$\text{or, } A = G\left[\frac{1}{i/m} - \frac{N\times m}{(1+i/m)^{N\times m} - 1}\right]$$

Book Numericals:

(5) 1> Find present worth of the following cash flow by using gradient method when rate of interest is 10% per year.

End of year	Net Cash Flow
1	2,00,000
2	1,75,000
3	1,50,000
4	1,25,000
5	1,00,000

Sol Given;

Equal Amount, A = Rs. 2,00,000
Growth rate, G = Rs. 25,000
Rate of interest, $i = 10\% = 0.1$
Time period, $N = 5$ years
Present worth, $P = ?$

We know,

$$P = P_A - P_G$$

$$= A\left[\frac{(1+i)^N - 1}{i(1+i)^N}\right] - \frac{G}{i}\left[\frac{(1+i)^N - 1 - Ni}{i(1+i)^N}\right]$$

$$= 200{,}000\left[\frac{(1+0.1)^5 - 1}{0.1(1+0.1)^5}\right] - \frac{25000}{0.1}\left[\frac{(1+0.1)^5 - 1 - 5\times 0.1}{0.1(1+0.1)^5}\right]$$

$$= 758157.354 - 171545.038$$

$\therefore P$ = Rs. 5,86,612.316

Hence, the present worth is Rs. 5,86,612.316. Ans

2015-Spring

Book

(37) 2> Ms sharma is planning for ~~her~~ retired life and has 10 more years ~~for~~ of service. She would like to deposit 20% of her salary, which is Rs. 4,000 at the end of first year and thereafter she wishes to deposit the amount with an annual increase of Rs. 1,000 for the next 9 years with an interest of 10%. Find the total amount at the end of 10th year.

Sol Given;

Time period, $N = 10$ years
Interest rate, $i = 10\% = 0.1$
Equal Amount, A = Rs. 4,000
Gradient rate, G = Rs. 1000
~~Present~~ Future worth, $F = ?$

We know,

$$F = F_A + F_G$$

$$\Rightarrow F = A\left[\frac{(1+i)^N - 1}{i}\right] + \frac{G}{i}\left[\frac{(1+i)^N - 1}{i}\right] - \frac{NG}{i}$$

$$= 4000\left(\frac{(1+0.1)^{10} - 1}{0.1}\right) + \frac{1000}{0.1}\left(\frac{(1+0.1)^{10} - 1}{0.1}\right) - \frac{10\times 1000}{0.1}$$

$$= 63749.698 + 159374.246 - 100000$$

$\therefore F$ = Rs. 123123.944

Hence, the Future worth after 10 years will be Rs. 1,23,123.944. Ans

4> Interest Calculation of Geometric gradient:

Let g = gradient rate

i> When g is increasing:

$$P_G = \frac{A_1}{i-g}\left[1-\left(\frac{1+g}{1+i}\right)^N\right] \qquad F_G = P_G\cdot(1+i)^N$$

where,

A_1 = Cashflow at the end of 1st year

ii> When g is decreasing:

$$P_G = \frac{A_1}{(1+g)}\left[1-\left(\frac{1-g}{1+i}\right)^N\right] \qquad F_G = P_G(1+i)^N$$

2018 - Spring

Numericals:

Book 29

1> Ramesh an Engineer is planning to place 20% of his salary, which is Rs. 2,50,000 per year present, each year in mutual fund. He expects 7% of his salary increase each year for next 15 years. If the mutual fund will average 10% annual return, what will be the sum-amount at the end of 15 years? If salary increases by Rs 25,000 per year, what will be the amount?

Soln Here;

Number of year, N = 15 years

~~Increment rate of salary = 7% per year = g~~

~~Present salary~~

Year	7% increment in salary per year	20% deposit	Rs 25,000 increment in salary /year	20% deposit
0	2,50,000	50,000	2,50,000	50,000
1	2,67,500	53,500	2,75,000	55,000
2	2,86,225	57,245	3,00,000	60,000
⋮	⋮	⋮	⋮	⋮
15	⋮	⋮	⋮	⋮

case I:

Present Amount, P = Rs. 50,000

Gradient rate, $g = 7\% = 0.07$

Annual return rate, $i = 10\% = 0.1$

Cashflow at the end of 1st year, A_1 = Rs. 53,500

Future Amount, F = ?

we know,

$$F = F_P + F_G = P(1+i)^N + P_G(1+i)^N$$

$$= P(1+i)^N + \frac{A_1}{i-g}\left[1-\left(\frac{1+g}{1+i}\right)^N\right](1+i)^N$$

$$= 50,000(1+0.1)^{15} + \frac{53,500}{0.1-0.07}\left[1-\left(\frac{1+0.07}{1+0.1}\right)^{15}\right](1+0.1)^{15}$$

$\therefore F =$ Rs. 27,38,015.39

Hence, the sum-amount at the end of 15 years is Rs. 27,38,015.39. Ans

case II:

Present Amount, P = 50,000

Gradeient rate, G = Rs. 5,000

Annual return rate, $i = 10\% = 0.1$

cashflow at the end of 1st year, A_1 = Rs. 55,000

Future Amount, F = ?

we know,

$$F = F_P + F_A + F_G$$

$$= P(1+i)^N + A_1\left[\frac{(1+i)^N-1}{i}\right] + \frac{G}{i}\left[\frac{(1+i)^N-1}{i}\right] - \frac{NG}{i}$$

$$= 50,000(1+0.1)^{15} + 55,000\left[\frac{(1+0.1)^{15}-1}{0.1}\right] +$$

$$\frac{5000}{0.1}\left[\frac{(1+0.1)^{15}-1}{0.1}\right] - \frac{15\times5000}{0.1}$$

$= 1956348.902 + 838624.0847$

$\therefore F =$ Rs. 27,94,972.987

Hence, the required amount will be Rs. 27,94,972.287. Ans

Book 5) Irregular / Uneven Cashflow Series:

(43)

① Calculate PW and FW form the following cash flow if MARR is 15% per year.

End of year	Net cash flow (Rs.000)
0	-1100
1	-550
2	0
3	230
4	1000
5	520
6	120
7	450

Sol: Here,

EoY	NCF	PF @15% $(1+i)^{-N}$	PW	PF @15% $(1+i)^{N}$	FW
0	-1100	1	-1100	2.66	-2926
1	-550	0.87	-478.5	2.31	-1270.5
2	0	0.76	0	2.01	0
3	230	0.66	151.8	1.75	402.5
4	1000	0.57	570	1.52	1520
5	520	0.50	260	1.32	686.4
6	120	0.43	51.6	1.15	138
7	450	0.38	171	1.00	450
			-374.1		-999.6

① If a bank charges 12% interest per month on housing loan. Find the effective rate of interest when compunding is
i) yearly ii) semi-annually iii) quarterly
iv) monthly v) Daily. vi) Hourly

Sol: Given;
Interest rate, $r = 12\%$ per month

i) Yearly:
No. of compounding period, $m = 1$
$\therefore i_{eff} = \left(1+\frac{r}{m}\right)^{m} - 1 = \left(1+\frac{0.12}{1}\right)^{1} - 1 = 0.12 = 12\%$

ii) Semi-Annually:
No. of compounding period, $m = 2$
$\therefore i_{eff} = \left(1+\frac{r}{m}\right)^{m} - 1 = \left(1+\frac{0.12}{2}\right)^{2} - 1 = 12.36\%$

iii) Quarterly:
No. of compounding period, $m = 4$
$\therefore i_{eff} = \left(1+\frac{r}{m}\right)^{m} - 1 = \left(1+\frac{0.12}{4}\right)^{4} - 1 = \cancel{0.1268}\ 0.1255 = 12.55\%$

iv) Monthly:
No. of compounding period, $m = 6$
$\therefore i_{eff} = \left(1+\frac{r}{m}\right)^{m} - 1 = \left(1+\frac{0.12}{6}\right)^{6} - 1 = 0.1268 = 12.68\%$

vi> Daily:

No. of compounding period, $m = 365$

$\therefore i_{eff} = \left(1+\frac{r}{m}\right)^{m} - 1 = \left(1+\frac{0.12}{365}\right)^{365} - 1 = 0.1275 = 12.75\%$

vii> Hourly:

No. of compounding period, $m = 8760$

$\therefore i_{eff} = \left(1+\frac{r}{m}\right)^{m} - 1 = \left(1+\frac{0.12}{8760}\right)^{8760} - 1 = \underline{12.74\%}$

4. Basic Methods of Engineering Economic Studies

4.1. MARR (Minimum Attractive Rate of Return):

→ MARR is the minimum interest rate that encourages the investor to invest in financial projects.

→ MARR is the interest rate at which a firm can always earn or borrow money under a normal operating environment.

→ MARR shows how much an investor is interested in investment.

→ MARR is determined from the opportunity cost view point.

4.2. Equivalence Worth Methods: PW, FW & AW

→ This method converts all cash flows into equivalent worth at present or future time by using an interest rate equal to MARR.

4.2.1. Present Worth (PW) method:

→ PW method is based on the concept of equivalent worth of all cash flows to the base point in time called the present.

→ It is also known as Net Present Value (NPV).

$$PW = -I + (R-E)\left[\frac{(1+i)^N - 1}{i(1+i)^N}\right] + S.V.\,(1+i)^{-N}$$

where,

I = Initial cost; S.V = salvage value;

R = Annual revenue or benefit

E = Annual cost or expenses

~~Decision criter~~

4.2.2. Future Worth (FW) method:

→ FW method is particularly used in an investment situation where we need to compute the equivalent worth of the project at the end of the investment period.

→ FW ~~method~~ is the equivalent worth of all cashflows at the end of study period.

$$FW = -I(1+i)^N + (R-E)\left[\frac{(1+i)^N - 1}{i}\right] + S.V$$

where;

I = Initial cost

R = Annual revenue or benefit

E = Annual cost or expenses

S.V. = salvage value

4.2.3. Annual Worth (AW) method:

→ AW method provides basis for measuring investment worth into a series of equal payments at the end of each period.

→ The AW of a project is annual equivalent revenue or savings minus equivalent expenses, less its annual capital recovery (CR) amount.

$$AW = R - E - \left[I\left(\frac{i(1+i)^N}{(1+i)^N - 1}\right) - S.V.\left(\frac{i}{(1+i)^N - 1}\right)\right]$$

where,

I = Initial cost

R = Annual revenue or benefits

E = Annual cost or expenses

S.V. = salvage value

Note: Decision Criteria;

→ PW, FW, AW > 0, accept the project

→ PW, FW, AW < 0, reject the project

→ PW, FW, AW = 0, remain indifferent.

Book

Numericals:

① Check the feasibility of the following investment by PW, FW and AW methods.

First investment (Rs.)	200,000
Salvage value (Rs.)	50,000
Project Period (Yrs)	10
Annual Revenue (Rs)	50,000
Annual Cost (Rs.)	15,000
MARR	10% per year

Sol. Given;

First Investment, I = Rs. 200,000

Salvage Value, S.V. = Rs. 50,000

Project Period, N = 10 yrs

Annual Revenue, R = Rs. 50,000

Annual cost, E = Rs. 15,000

MARR, i = 10% per year

We know,

i) $PW = -I + (R-E)\left[\frac{(1+i)^N - 1}{i(1+i)^N}\right] + S.V(1+i)^{-N}$

$= -2,00,000 + (50,000 - 15,000)\left[\frac{(1+0.1)^{10} - 1}{0.1(1+0.1)^{10}}\right]$

$+ 50,000(1+0.1)^{-10}$

$\therefore PW = Rs. 34,337.01 > 0$

Since, PW(10%) > 0, so the project is feasible. Ans

ii) $FW = -I(1+i)^N + (R-E) \times \left[\frac{(1+i)^N - 1}{i}\right] + S.V.$

$= -200000(1+0.1)^{10} + (50,000 - 15,000) \times \left[\frac{(1+0.1)^{10} - 1}{0.1}\right]$

$+ 50,000$

$\therefore FW = Rs. 89,061.37 > 0$

Since, FW(10%) > 0, so the project is feasible. Ans

iii) $AW = (R-E) - \left[I\left(\frac{i(1+i)^N}{(1+i)^N - 1}\right) - S.V.\left(\frac{i}{(1+i)^N - 1}\right)\right]$

$= (50,000 - 15000) - \left[200,000\left(\frac{0.1(1+0.1)^{10}}{(1+0.1)^{10} - 1}\right) - 50,000\left(\frac{0.1}{(1+0.1)^{10} - 1}\right)\right]$

$= Rs. 5,588.19 > 0$

Since, (AW) > 0, so the project is feasible. Ans

BOOK

(2) 2> Evaluate the following project on the basis of the PW and FW methods when the MARR is 10 %.

First Investment (Rs.)	1,50,000
Salvage Value (Rs.)	35,000
Project Period (~~Rs.~~)	15
Annual expenses (Rs.)	10,000
Annual revenue (Rs.)	40,000
Overhead cost (Rs.) at the end of 5th yr	25,000
Overhead cost (Rs.) at the end of 10th yr	50,000

Soln Here,

First Investment, I = Rs. 1,50,000

Salvage Value, S.V. = Rs. 35,000

Project Period, N = 15 yrs

Annual expenses, E = Rs. 10,000

Annual revenue, R = Rs. 40,000

Overhead cost at the end of 5th year, = Rs. 25,000

Overhead cost at the end of 10th year, = Rs. 50,000

We know,

i) $PW = -I + (R-E)\boxed{\frac{(1+i)^N - 1}{i(1+i)^N}} + S.V.(1+i)^{-N}$

$- O.C.(5)(1+i)^{-5} - O.C.(10)(1+i)^{-10}$

$= -1,50,000 + (40,000 - 10,000)\boxed{\frac{(1+0.1)^{15} - 1}{0.1(1+0.1)^{15}}}$

$+ 35,000(1+0.1)^{-15} - 25000(1+0.1)^{-5} - 50,000(1+0.1)^{-10}$

∴ PW(10%) = Rs. 51,760.9 Ans

ii) $FW = -I(1+i)^N + (R-E)\boxed{\frac{(1+i)^N - 1}{i}} + S.V\cancel{(1+i)^{-N}}$

$- O.C(5)(1+i)^{-5} - O.C(10)(1+i)^{-10}$

$= -1,50,000(1+0.1)^{15} + (40,000 - 10,000)\boxed{\frac{(1+0.1)^{15} - 1}{0.1}}$

$+ 35,000\cancel{(1+0.1)^{-15}} - 25,000(1+0.1)^{-5} - 50,000(1+0.1)^{-1}$

~~= 334965.9471 − 34800.1976~~

∴ ~~FW = Rs.~~

= −626587.2254 + 953174.4508 − 34800.1976

+ 35000

= Rs. 326986.8302

∴ FW(10%) = Rs. 3,26,986.8302. Ans